CLASS 7 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A man goes from A to B at 4 km/h and returns at 6 km/h. The time taken in the round trip is 5 h. The distance AB is: 1

(a) 6 km **(b)** 5 km **(c)** 12 km **(d)** 24 km

2. In one hour this volume is: 1

(a) 8 litres **(b)** 480 litres **(c)** 4800 litres **(d)** 48 litres

3. Which of these statements about menarche is correct? 1

(a) it can occur from about 10 to 15 years **(b)** it occurs at 60 years
(c) it never occurs **(d)** it occurs at 5 years

4. Which of the following is a correct application of radiation? 1

(a) solar cooker **(b)** melting of ice **(c)** heating of milk **(d)** spoon in tea

5. A student measures one oscillation of a pendulum with a stopwatch of reading to 0.1 s and gets 2.0 s. The error is about: 1

(a) 50 per cent **(b)** up to 5 per cent
(c) up to 0.5 per cent **(d)** nil

6. A carnivore's digestive tract compared to a herbivore's is: 1

(a) same **(b)** absent **(c)** shorter **(d)** longer

7. A nurse says that the thyroid gland needs iodine. A diet lacking it may lead to: 1

(a) swelling of the neck **(b)** swelling of the foot
(c) swelling of the arm **(d)** swelling of the eye

8. A student places a thermometer in sunlight and another in shade. The one in sunlight reads higher due to: 1

(a) conduction **(b)** radiation **(c)** convection **(d)** evaporation

9. His displacement after one full round is: 1

(a) 0 **(b)** 100 m **(c)** 4 m **(d)** 400 m

10. The villi are absent in the intestine. The consequence is: 1

(a) no excretion **(b)** no breathing
(c) no digestion **(d)** poor absorption of digested food

11. If a person's pancreas stops making insulin, blood sugar will: 1

(a) rise **(b)** stay same
(c) become acidic always **(d)** fall to zero

12. In deserts, nights are cold and days are hot because sand: 1

(a) reflects all **(b)** heats slowly
(c) heats and cools quickly **(d)** stores heat

13. A boat sails 24 km in 3 h with a steady speed. In 5 h it travels: 1

(a) 8 km **(b)** 120 km **(c)** 40 km **(d)** 15 km

14. A blood sample has 5 million RBC per mm^3. In 2 mm^3 the number is: 1

(a) 10 million **(b)** 7 million **(c)** 2.5 million **(d)** 5 million

15. A girl's period is irregular during the first year. This is: 1

(a) a danger **(b)** rare
(c) common as cycles settle **(d)** a disease

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why is acceleration not required to solve problems on uniform speed? 2

17. Why does oxygen move from the alveoli to blood? 2

18. Why do two persons of the same age show different features of puberty? 2

19. Why are the fins of a radiator painted black and made of metal? 2

20. A student recorded: 10 oscillations in 15.2 s, 15.0 s, 15.4 s. Find the mean time period. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A person goes from A to B at 40 km/h and returns at 60 km/h. Find the average speed. 3

22. A runner's pulse rate goes from 70 to 150 per minute. Explain what happens in the muscles and the blood. 3

23. Explain the importance of exercising in daylight for bones. 3

24. Explain how a refrigerator removes heat from inside to outside (in simple words). 3

OR

Explain the use of the 'seconds pendulum' in old clocks. 3

25. Why does a diabetic person need to be careful about sugar-rich foods? 3

OR

How would you respond if someone teases a classmate about body changes? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Compare the roles of nervous system and endocrine system in adolescent changes and in responding to stress. 5

OR

Explain, with a labelled description, how a solar cooker works and why each part is designed as it is. 5

27. Design an experiment to show how the time period of a pendulum varies with its length. Specify measurements, tables and expected relation. 5

OR

Describe the flow of blood through the double circulation: right heart to lungs, lungs to left heart, left heart to the body. Explain why this is called double circulation. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND